1. $m=3$, $P(3,-1)$; point-slope: $y+1=3(x-3)$, slope-intercept: $y=3x-10$

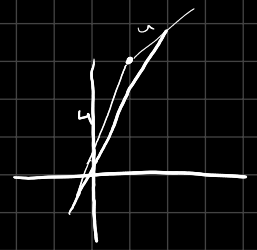
9. $m=-5$, $P(\sqrt{3},2\sqrt{3})$; point-slope: $y-2\sqrt{3}=-5(x-\sqrt{3})$, slope-intercept: $y=-5x+7\sqrt{3}$

11. $P(0,0)$, $Q(-3,5)$; $m=\frac{\Delta y}{\Delta x}=\frac{5-0}{-3-0}=-\frac{5}{3}$ slope-intercept: $y=-5/3\,x$

19. $P(\sqrt{2},-\sqrt{2})$, $Q(-\sqrt{2},\sqrt{2})$; $m=\frac{\sqrt{2}-(-\sqrt{2})}{-\sqrt{2}-\sqrt{2}}=\frac{2\sqrt{2}}{-2\sqrt{2}}=-1$, slope-intercept: $y=-x+2\sqrt{2}$

21. $f(x)=2x-1$, $m=2$, y-intercept: -1, x-intercept: $1/2$

27. $y=2x+1$, $P(-1,3)$;



$$4=\sqrt{(x_1-x_0)^2+(y_1-y_0)^2}=\sqrt{(x_1+1)^2+(y_1-3)^2}$$
$$=\sqrt{(x+1)^2+(2x+1-3)^2}=\sqrt{x^2+2x+1+4x^2-8x+4}$$
$$=\sqrt{5x^2-6x+5}$$
$$16=5x^2-6x+5$$
$$5x^2-6x-11=0,\quad x_1=11/5,\quad x_2=-1,\quad y_1=\frac{27}{5},\quad y_2=-1$$

44. $f(x)=x^3$, $[-1,2]$, $\frac{\Delta y}{\Delta x}=\frac{8-(-1)}{2-(-1)}=3$

50. $f(x)=x^3$, $[x,x+h]$, $\frac{\Delta y}{\Delta x}=\frac{(x+h)^3-x^3}{x+h-x}=\frac{x^3+3x^2h+3xh^2+h^3-x^3}{h}=\frac{3x^2h+3xh^2+h^3}{h}=3x^2+3xh+h^2$

60. $y=-6x+5$, $P(3,2)$; $y_2=-6x+20$

67. $y=\frac{2}{3}x-7$, $P(6,0)$; $y_2=-\frac{3}{2}x+9$